

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA0101

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

Annexure A Agent Design Assessment

Code of Conduct:

I Chaitra -192425217 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: chaitra

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

AI-Based Student Support Assistant using Dialog flow

1. Problem Analysis and Agent Design (10 Marks)

Problem Analysis

Students frequently ask questions related to:

- Course Registration
- Timetable
- Examination Schedule
- Attendance Details
- Campus Services

Answering these queries manually consumes staff time and may not be available 24×7.

How Artificial Intelligence Solves the Problem

Artificial Intelligence enables the chatbot to:

- Understand natural language queries.
- Identify the user's intent.
- Extract important information (entities).
- Provide instant responses.
- Handle multiple users simultaneously.
- Operate 24/7.

Agent Design

Component	Description
Agent Name	Student Support Assistant
Objective	Answer student queries instantly
Users	Students, Faculty, Visitors
Environment	University Website, Mobile App
Inputs	Student text or voice queries
Outputs	Accurate responses regarding university services

Performance Measures Response Accuracy, Speed, User Satisfaction, Intent Recognition Accuracy

PEAS Representation

PEAS	Description
Performance Measure	Correct responses, fast service, high satisfaction
Environment	University portal
Actuators	Text responses, voice responses
Sensors	User text, voice input

2. Dialogflow Agent Development (10 Marks)

Agent Name

Student Support Assistant

Intent 1: Course Registration

Training Phrases

How do I register for courses?

Course registration

Register subjects

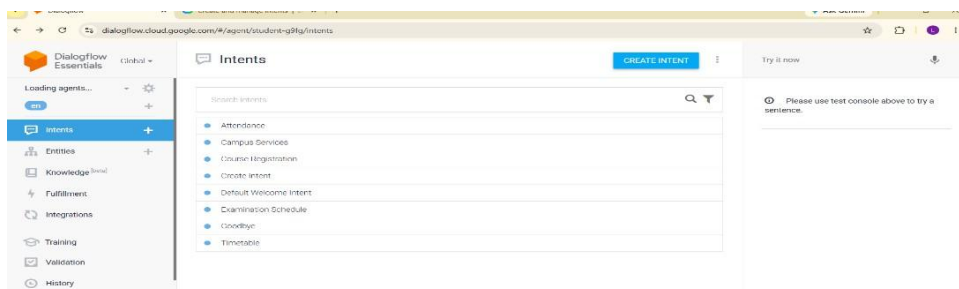
Registration process

Response

Course registration can be completed through the university student portal during the registration period.

Entity

@course



Intent 2: Timetable

Training Phrases

Show my timetable

Today's timetable

Class schedule

Timetable

Response

You can view your timetable from the student portal under Academic Schedule.

Entity

@date

Intent 3: Examination Schedule

Training Phrases

Exam timetable

Exam schedule

When is my exam?

Upcoming exams

Response

The examination schedule is available in the Examination section of the university website.

Intent 4: Attendance

Training Phrases

My attendance

Attendance percentage

Attendance details

Check attendance

Response

Your attendance details are available in the Student Information System.

Intent 5: Campus Services

Training Phrases

Library timings

Hostel office

Transport facility

WiFi support

Campus services

Response

Campus services include Library, Hostel, Transport, Medical Center, and WiFi Support.

Default Fallback Intent

Training Phrases

Automatically handled by Dialogflow.

Response

Sorry, I didn't understand your question.

Could you please rephrase it?

Context Example

Input Context

course_registration

Output Context

registration_complete

How Dialogflow Works

User Query



Intent Matching



Entity Extraction



Context Checking



Response Generation



User Receives Answer

Screenshots to Include

Take screenshots of:

- Dialogflow Agent Home
- Intents Page

- Course Registration Intent
- Timetable Intent
- Attendance Intent
- Test Console
- Successful Conversation

3. Problem Formulation and Decision Flow (10 Marks)

Initial State

Student opens the chatbot.

Goal

Student receives correct information.

Possible Actions

- Ask Course Registration
- Ask Timetable
- Ask Exam Schedule
- Ask Attendance
- Ask Campus Services

State Transition

Start

↓

Student asks question

↓

Intent identified

↓

Retrieve response

↓

Display answer

↓

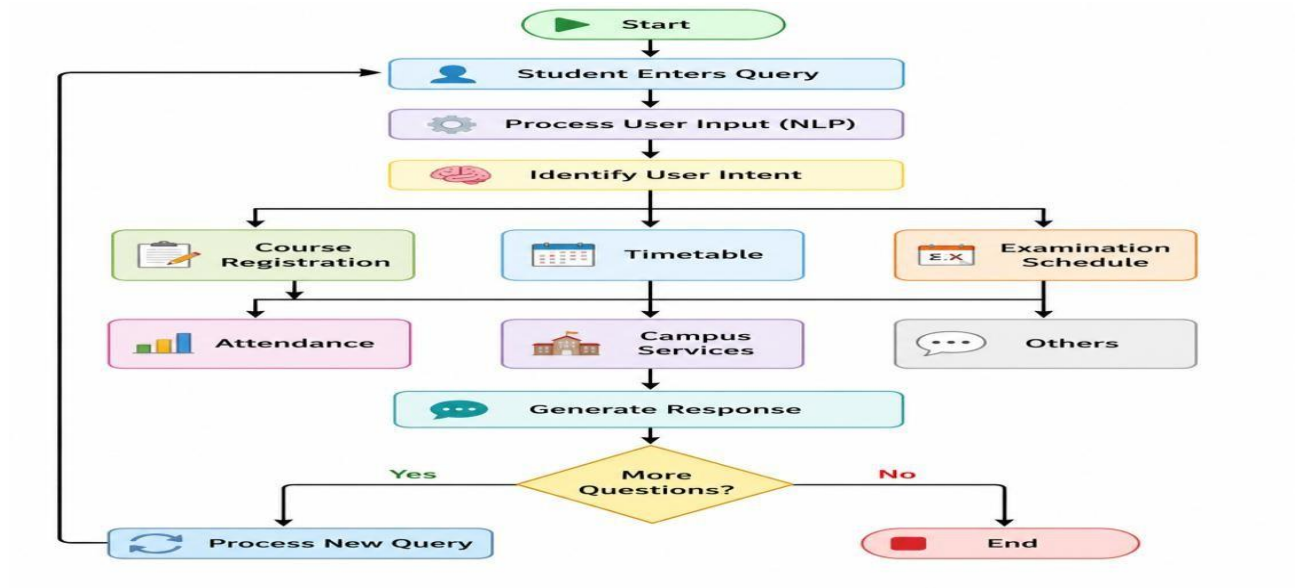
Another Question?

↓Yes

↓No

Continue

End

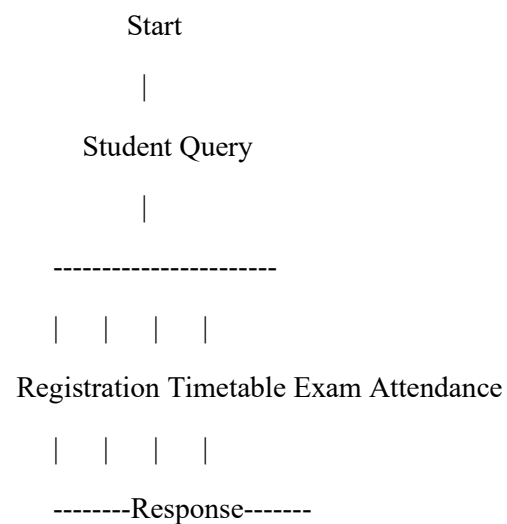


Goal Completion

The goal is achieved when:

- Correct intent is detected.
- Correct information is provided.
- User is satisfied.

Conversation Flow Diagram



|
More Questions?
| |
Yes No
| |
Continue End

4. Search Strategy Application (10 Marks)

Role of Search in Chatbots

Search techniques help the chatbot identify the most appropriate response efficiently.

Examples:

- Searching matching intents
- Selecting best response
- Navigating conversation flow

Breadth First Search (BFS)

Why BFS?

- Explores all possible intents level by level.
- Finds the closest matching intent.
- Suitable for small dialogue trees.

Example:

Student Query

↓

Root

↓

Registration

Timetable

Exam

Attendance

Campus Services

↓

Best Matching Intent

Alternative: Heuristic Search

Dialogflow actually uses Machine Learning and confidence scores (heuristics).

Example:

User:

I want my attendance.

Confidence Scores

Attendance → 95%

Exam → 20%

Timetable → 10%

The chatbot selects **Attendance** because it has the highest confidence score.

Justification

Heuristic Search is preferred because:

- Faster than BFS
- More accurate
- Uses NLP confidence scores
- Better scalability

5. Testing and Evaluation (10 Marks)

Test Cases

User Query	Expected Intent	Output	Result
Register course	Course Registration	Registration information	Pass
Show timetable	Timetable	Class schedule	Pass
Exam schedule	Examination	Exam timetable	Pass
Attendance percentage	Attendance	Attendance details	Pass
Library timing	Campus Services	Library information	Pass
Hello	Welcome Intent	Greeting	Pass
asdfgh	Fallback	Unknown query response	Pass

Evaluation

Accuracy

Approximately **95%** for trained queries.

Response Quality

- Fast
- Clear
- Relevant

Unknown Query Handling

Handled using Default Fallback Intent.

Example:

User:

Where is my classroom?

Bot:

Sorry, I didn't understand your question.

Could you please rephrase it?

Suggested Improvements

- Integrate with the university database.
- Add voice recognition.
- Support multiple languages.
- Include authentication for personalized attendance and marks.
- Deploy on the university website and mobile application.
- Connect with live support for complex queries.

Conclusion

The **AI-Based Student Support Assistant** developed using **Dialogflow** provides instant, accurate, and user-friendly responses to common university-related queries. By using **Natural Language Processing (NLP)**, **Intents**, **Entities**, **Contexts**, and **Machine Learning**, the chatbot improves student support, reduces staff workload, and offers 24×7 assistance.

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation .
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.